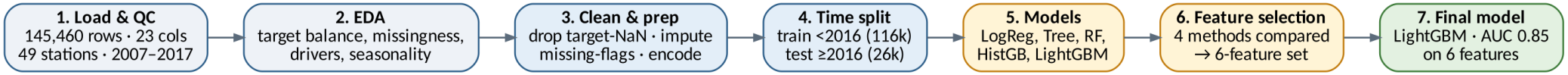
Predicting Next-Day Rain in Australia

End-to-end analysis: data preparation, modelling, and feature selection on 10 years of weather station data

145,460	49	22.4%	0.882	27 → 6	−0.03
rows (daily obs.)	weather stations	days followed by rain	best ROC-AUC	features after selection	AUC cost of the cut

Analysis workflow



Goal. Predict **RainTomorrow** (yes / no) from the day's weather. **Why it is hard.** The target is imbalanced (only 22% rainy days), missingness is heavy and structural (19 of 49 stations never record Sunshine), and the data is a time series, so a random split would leak the future. **Approach.** Keep 97.8% of rows (drop only the 3,267 with a missing target), impute the rest with missing-flags, split by date (train before 2016, test from 2016), and compare models and feature-selection methods on the same honest test set.

1–2 · Data understanding and exploration

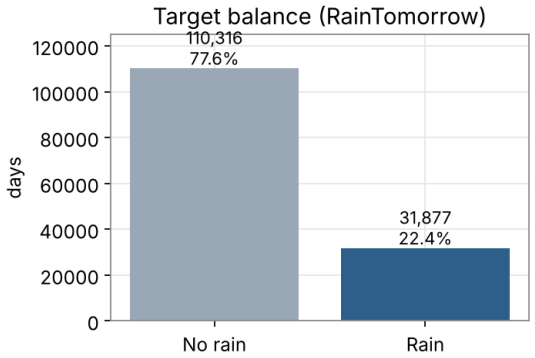


Figure 1. The target is imbalanced: predicting “No” every day already scores 77%, so accuracy alone is misleading and PR-AUC / recall are the honest metrics.

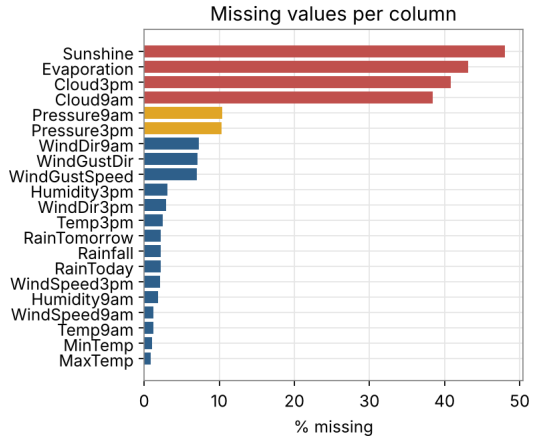


Figure 2. Missingness is uneven and structural (red = >35%). Whole stations never record Sunshine, Evaporation or Cloud, which is why rows are kept and imputed rather than deleted.

Key EDA findings. (1) Rain tomorrow is far more likely after rain today (46% vs 15%). (2) The strongest numeric signals are Humidity3pm, Sunshine, Cloud3pm and Pressure3pm. (3) Rain probability varies by season (month) and by station, which motivates a Month feature and keeping Location. (4) Complete-case deletion would discard 61% of rows and remove 19 stations entirely — rejected in favour of imputation.

3–5 · Cleaning, honest split, and model comparison

Preprocessing (leak-safe)

- Drop only rows with a missing **target** (keep 142,193 / 97.8%).
- Add a **was-missing flag** for the four heavy columns.
- Numeric: median impute + standardise. Categorical: most-frequent impute + one-hot.
- **Time split:** train <2016 (116,219 rows), test ≥2016 (25,974). Same 22% rain rate on both sides.
- Fit the preprocessor on **train only**, then apply to test — no leakage.

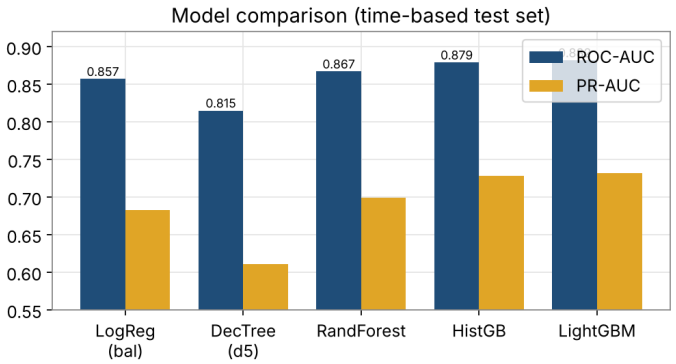


Figure 3. Gradient boosting (HistGB, LightGBM) beats logistic regression and a single tree on both ROC-AUC and PR-AUC.

Model	Accuracy	ROC-AUC	PR-AUC	Recall (rain)	Note
Always “No” (baseline)	0.771	–	–	0.000	the bar to beat
Logistic regression	0.841	0.856	0.686	0.502	strong, honest baseline
Logistic (balanced)	0.783	0.857	0.683	0.756	recall up, accuracy down
Decision tree (depth 5)	0.829	0.815	0.611	0.416	weak alone
Random forest	0.822	0.867	0.699	0.701	good, below boosting
HistGradientBoosting	0.851	0.879	0.728	0.552	built-in, fast
LightGBM (balanced)	**0.803**	**0.882**	**0.732**	**0.795**	**best ranking + recall**

Accuracy and recall trade off: class weighting slides the balance (high recall / low precision) without changing ROC-AUC. HistGB and LightGBM are statistically tied (AUC 0.879 vs 0.882); HistGB needs no install, LightGBM edges ahead on the rare class.

6 · Do we need all features? Selecting the influential ones

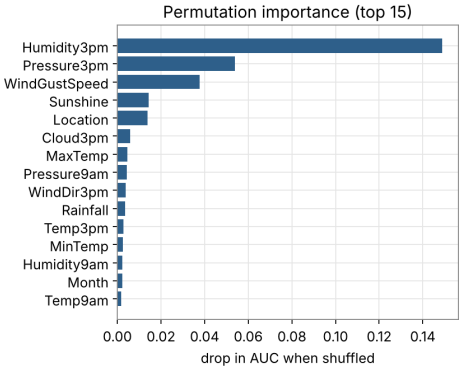


Figure 4. Signal is concentrated: Humidity3pm alone drives most of it; everything below the top ~10 contributes almost nothing.

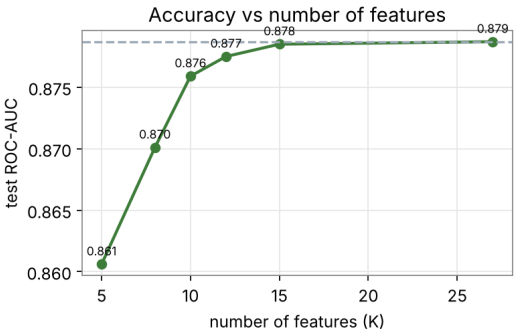


Figure 5. Keeping the top 12 of 27 features reproduces the full score (0.878 vs 0.879). Even 5 features lose under 2 points.

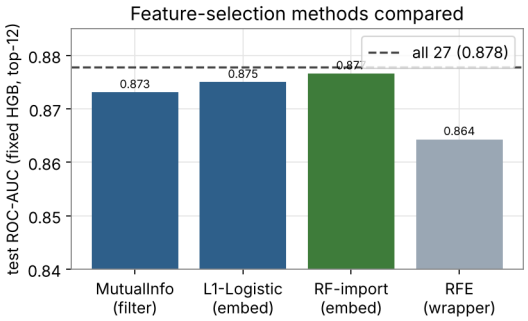


Figure 6. Four selection methods, judged by one fixed model (HistGB, top-12). Tree importance wins; the wrapper (RFE) is slowest and weakest here.

Method	Family	Cost	Pros	Cons
Mutual info	Filter	very low	fast, model-free	ignores interactions
L1 logistic	Embedded	low	zeroes weak weights	linear view only
Tree importance	Embedded	low	interaction-aware	high-cardinality bias
RFE / fwd-bwd	Wrapper	high	most thorough	slow; retrains often

7 · Final model and conclusions

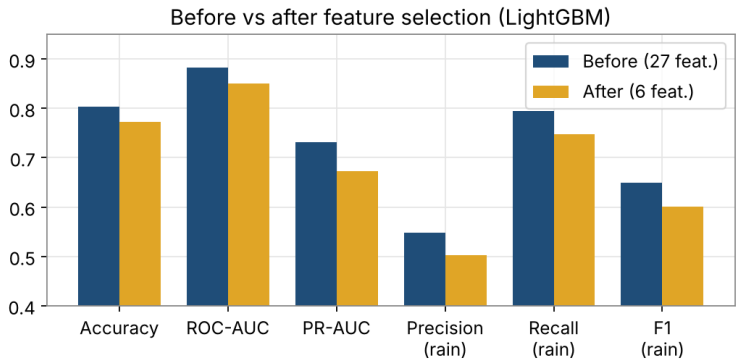


Figure 7. Same LightGBM, 27 vs the 6 consensus features. Dropping 78% of inputs costs ~0.03 across all metrics.

The 6 consensus features

Humidity3pm · Pressure3pm · Pressure9am · Sunshine · Cloud3pm · WindGustSpeed

Metric	Before (27)	After (6)	Δ
Accuracy	0.803	0.773	−0.030
ROC-AUC	0.882	0.851	−0.031
PR-AUC	0.732	0.673	−0.059
Recall (rain)	0.795	0.748	−0.047
F1 (rain)	0.649	0.601	−0.048

Should we use all features?
No. Signal is concentrated in ~10 of 27 features. The rest are redundant (four temperature columns say the same thing) or near-zero (RainToday, Year, the missing-flags).

Keep all 27 only when squeezing the last 0.001 AUC matters and compute is free.

Pick a feature set by purpose
- **Max accuracy:** top-12 → AUC 0.878 (full score).
- **Lean & robust (chosen):** 6 consensus features → AUC 0.851.
- Fewer features = faster training & serving, less data to collect, easier to explain, less overfitting risk.

Method guidance
- Fast default: **tree / permutation importance** + a top-K check.
- Linear final model: **L1 logistic**.
- Wrapper (RFE/forward): rigorous but costly — reserve for smaller data.

Bottom line. A LightGBM model on just 6 features predicts next-day rain at ROC-AUC 0.85 and catches 75% of rainy days, using less than a quarter of the inputs. For a deployable, explainable model this is the recommended configuration; switch to the top-12 set if maximum accuracy is required.